The Proof of Penrose's Singularity Theorem

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7/18/2026

1 The theorem

We will use the following standard form of the theorem.

Theorem 1 (Penrose)**.** *Let $(\mathcal{M}, g)$ be a time oriented spacetime with a noncompact Cauchy hypersurface Σ. Suppose that $R_{ab}N^aN^b \geq 0$ for every null vector N, and that Σ contains a trapped surface T^2. Then $(\mathcal{M}, g)$ is not future null geodesically complete.*

By Einstein's field equation $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$, contraction with a null vector N eliminates both terms proportional to the metric and gives $R_{\mu\nu}N^\mu N^\nu = \kappa T_{\mu\nu}N^\mu N^\nu$. Thus, for $\kappa > 0$, the curvature condition in the theorem is precisely the *null energy condition* (NEC), $T_{\mu\nu}N^\mu N^\nu \geq 0$.

The NEC follows from the stronger *dominant energy condition* (DEC), a natural physical assumption saying that, for every future-directed causal vector V, the energy current $\mathcal{J}^\mu = -T^\mu{}_\nu V^\nu$ is itself future-directed and causal. Take $V = N$ to be future-directed and null. Since two future-directed causal vectors have nonpositive inner product in the $(-,+,+,+)$ convention, $g(\mathcal{J}, N) \leq 0$. Consequently $T_{\mu\nu}N^\mu N^\nu = -g(\mathcal{J}, N) \geq 0$, which is the NEC. (If the DEC is stated only for timelike V, the same conclusion follows by approximating N by future-directed timelike vectors and taking a limit.) Physically, the DEC is natural because it requires the measured energy density to be nonnegative and the associated energy flux not to propagate faster than light. Our theorem only needs its weaker null consequence.

2 Definitions

Definition 1. *A spacetime is a connected time-oriented Lorentzian manifold $(\mathcal{M}, g)$ of dimension four and signature $(-,+,+,+)$.*

Definition 2. *A Cauchy hypersurface $\Sigma \subset \mathcal{M}$ is a spacelike manifold of the form $\Sigma = f^{-1}(c)$, where $f : \mathcal{M} \to \mathbb{R}$ is a smooth function with $c \in \mathbb{R}$ being its regular value, and Σ meets every inextendible timelike curve exactly once.*

Thus a Cauchy hypersurface supplies a global synchronization slice. Every inextendible timelike observer crosses the same slice exactly once.

Definition 3. *Given a spacetime $\mathcal{M}$ with Cauchy hypersurface Σ, a trapped surface $T^2 \subset \Sigma$ is a compact orientable two-dimensional submanifold for which both future null expansions θ_+ and θ_- are negative.*

The expansion θ will be defined below. Intuitively, it is the infinitesimal fractional area-growth rate of a light wavefront emitted from the trapped surface.

Definition 4. *A null geodesic γ is* complete *if its maximal affine-parameter domain is $\mathbb{R}$. It is* future complete *if that domain is unbounded above. A spacetime is future null geodesically complete if every future-directed null geodesic is future complete.*

3 Analytical Aspects

3.1 Gaussian Null Coordinates

Let ν be a unit spacelike normal to T^2 within Σ. Let T be the future-directed unit normal to Σ along T^2. The two normalized future null normals are

$$N = \frac{T+\nu}{\sqrt{2}}, \qquad W = \frac{T-\nu}{\sqrt{2}}, \qquad g(N,W) = -1.$$

We use N for the outward direction and W for the inward direction.

Define the two normal exponential maps

$$F(p,\lambda) = \exp_p(\lambda N_p), \qquad G(p,\lambda) = \exp_p(\lambda W_p).$$

For sufficiently small $t > 0$, their images $\mathcal{X} = F(T^2 \times (0,t))$ and $\mathcal{Y} = G(T^2 \times (0,t))$ are immersed null hypersurfaces. We carry out the calculation on $\mathcal{X}$. The calculation on $\mathcal{Y}$ is identical. Choose coordinates (y_1, y_2) on T^2 and use λ as the affine parameter along the generators. Parallel transport W along the curves $\lambda \mapsto F(p,\lambda)$ and exponentiate in the W direction to obtain Gaussian null coordinates (y_1, y_2, λ, r) near $\mathcal{X}$. In these coordinates $N = \partial_\lambda$, $W = \partial_r$, and $J_i = \partial_{y_i}$ are the transverse Jacobi fields. Thus

$$T_p\mathcal{M} = \langle W, N, J_1, J_2 \rangle.$$

The wavefront at affine parameter λ is $\mathcal{X}_\lambda = F(T^2 \times \{\lambda\})$. If u is a local optical function with $\mathcal{X} = \{u = 0\}$ and $N^a = g^{ab}(du)_b$, then $J_i \in T_p\mathcal{X}_\lambda$ gives $g(N, J_i) = du(J_i) = 0$. The construction also gives $g(W, J_i) = 0$ and preserves $g(N,W) = -1$ along $\mathcal{X}$. Define the screen endomorphism $\widehat{B} : T\mathcal{X}_\lambda \to T\mathcal{X}_\lambda$ by $\widehat{B}(J) = \nabla_J N = \nabla_N J$. Indeed,

$$\begin{aligned} 0 =& \nabla_N g(N, J_i) = g(N, \nabla_N J_i) \\ 0 =& \nabla_N g(W, J_i) = g(W, \nabla_N J_i) \end{aligned}$$

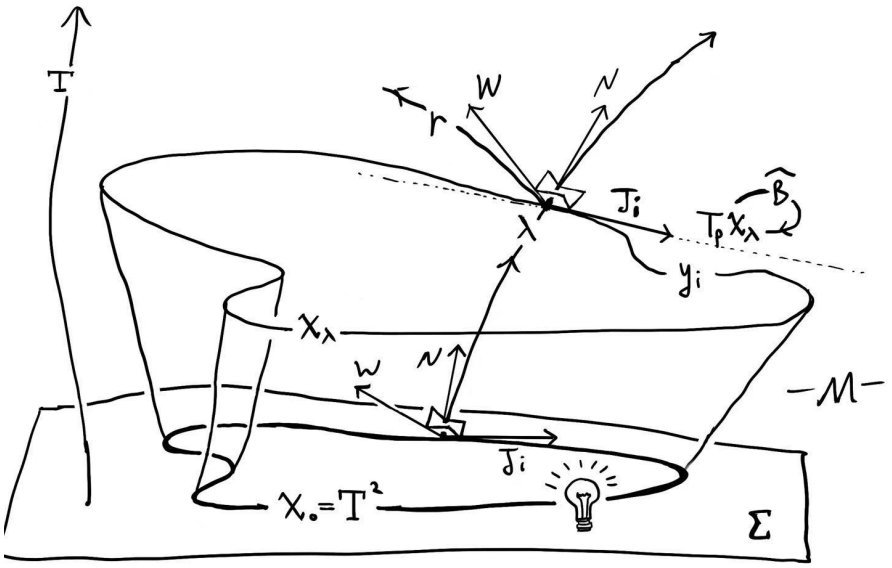


Figure 1: Gaussian null coordinate construction

so $\nabla_N J \in T\mathcal{X}_\lambda$ for every $J \in T\mathcal{X}_\lambda$. The null expansion is $\theta = \mathrm{tr}(\widehat{B})$. Denote the initial expansion of the outward F-family by θ_+ and that of the inward G-family by θ_-. Set

$$\theta_0 = \max_{p\in T^2}\{\theta_+(p), \theta_-(p)\} < 0.$$

At the end of this section, we will show $\theta \to -\infty$ before λ hits $\frac{2}{|\theta_0|}$. The Gaussian null coordinates will no longer provide an immersion beyond the first focal point. However, the maps F and G remain meaningful wherever the corresponding null geodesics exist.

Lemma 1. *If E_1, E_2 is an orthonormal basis of $T_p\mathcal{X}_\lambda$, then*

$$\theta = \sum_{A=1}^{2} g(\nabla_{E_A} N, E_A).$$

Proof. On $T\mathcal{X}_\lambda$, the endomorphism $J \mapsto \nabla_J N$ is precisely $\widehat{B}$. Taking its trace in an orthonormal basis gives the displayed formula. □

Lemma 2.

$$g|_{r=0} = -2\,dr\,d\lambda + h_{ij}\,dy^i\,dy^j.$$

Here h_{ij} is the induced Riemannian metric on $\mathcal{X}_\lambda$. If $h = \det(h_{ij})$, then $\partial_\lambda\sqrt{h} = \theta\sqrt{h}$.

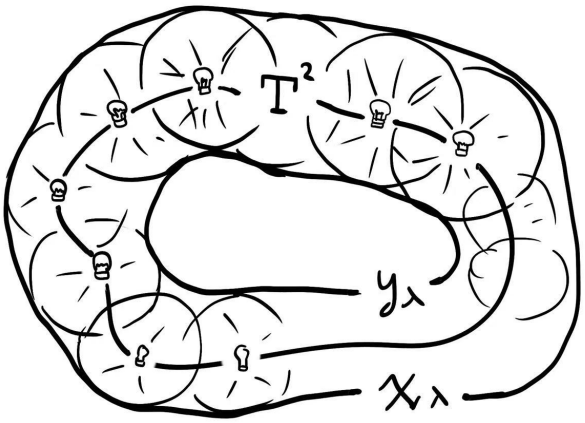


Figure 2: $\mathcal{X}_\lambda, \mathcal{Y}_\lambda$ as wave fronts

Proof. The form of $g|_{r=0}$ follows from the Gaussian null-coordinate construction and the normalization $g(N, W) = -1$. Since $N = \partial_\lambda$,

$$\theta = \Gamma^i_{i\lambda} = \frac{1}{2} g^{i\mu} \left(\partial_\lambda g_{\mu i} + \partial_i g_{\mu\lambda} - \partial_\mu g_{i\lambda} \right).$$

where $i, \mu \in \{1, 2\}$. Since $g_{i\lambda} = 0$ and $g_{ij} = h_{ij}$ on $\mathcal{X}_\lambda$, this gives $\theta = \frac{1}{2} h^{ij} \partial_\lambda h_{ij}$. Apply the identity

$$\frac{\partial h}{\partial \lambda} = h\, h^{ij} \frac{\partial h_{ij}}{\partial \lambda}$$

Therefore,

$$\frac{\partial \sqrt{h}}{\partial \lambda} = \frac{1}{2\sqrt{h}} \frac{\partial h}{\partial \lambda} = \frac{1}{2\sqrt{h}}\, h\, h^{ij} \frac{\partial h_{ij}}{\partial \lambda} = \frac{1}{2} \sqrt{h}\, h^{ij} \frac{\partial h_{ij}}{\partial \lambda} = \theta \sqrt{h}$$

□

Hence θ is the local fractional rate of change of transverse area.

Definition 5. *The* screen projection *along N and W is $P^a{}_b = \delta^a{}_b + N^a W_b + W^a N_b$. Its image is $T_p \mathcal{X}_\lambda$, and $P^2 = P$.*

Lemma 3. *Let $B^a{}_b = \nabla_b N^a$. Then $\nabla_N P = 0$ and $\theta = B^a{}_b P^b{}_a$.*

Proof. The identities $\nabla_N N = 0$ and $\nabla_N W = 0$ imply $\nabla_N P = 0$. Since P is the identity on $T_p \mathcal{X}_\lambda$, the restriction of $B \circ P$ to the screen is $\widehat{B}$. Taking the trace gives $\theta = B^a{}_b P^b{}_a$. □

Remark 1. *On the two-dimensional screen, $\widehat{B} = \frac{1}{2}\theta I + \widehat{\omega} + \widehat{\sigma}$, where $\widehat{\omega}$ is antisymmetric and $\widehat{\sigma}$ is symmetric and traceless.*

Lemma 4 (Raychaudhuri's equation)**.**

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \widehat{\sigma}^{ab}\widehat{\sigma}_{ab} + \widehat{\omega}^{ab}\widehat{\omega}_{ab} - R_{ab}N^aN^b$$

Proof. From the definition of θ, and since $\nabla_N P = 0$, we have

$$\begin{aligned}\frac{d\theta}{d\lambda} &= \nabla_N \left(B^a_b P^b_a\right)\\ &= P^b_a \nabla_N B^a_b\\ &= P^b_a N^c \nabla_c \nabla_b N^a.\end{aligned}$$

Now use the definition of the Riemann curvature tensor,

$$\begin{aligned}\frac{d\theta}{d\lambda} &= P^b_a N^c \left(\nabla_b \nabla_c N^a + R^a_{dcb} N^d\right)\\ &= P^b_a \left[\nabla_b \left(N^c \nabla_c N^a\right) - \left(\nabla_b N^c\right)\left(\nabla_c N^a\right)\right] + P^b_a R^a_{dcb} N^c N^d\\ &= -B^c_b P^b_a B^a_c - R_{cd}N^cN^d.\end{aligned}$$

Since P restricts to the identity on the screen and $\widehat{B}$ is the screen restriction of B, the quadratic term is $B^c{}_b P^b{}_a B^a{}_c = \widehat{B}^c{}_a \widehat{B}^a{}_c$. Therefore

$$\frac{d\theta}{d\lambda} = -\widehat{B}^c{}_a \widehat{B}^a{}_c - R_{ab}N^aN^b$$

For the first term, we have

$$\begin{aligned}\widehat{B}^c{}_a \widehat{B}^a{}_c &= \left(\frac{1}{2}\theta P^c_a + \widehat{\sigma}^c{}_a + \widehat{\omega}^c{}_a\right)\left(\frac{1}{2}\theta P^a_c + \widehat{\sigma}^a{}_c + \widehat{\omega}^a{}_c\right)\\ &= \frac{1}{4}\theta^2 P^c_a P^a_c + \widehat{\sigma}^c{}_a \widehat{\sigma}^a{}_c + \widehat{\omega}^c{}_a \widehat{\omega}^a{}_c.\end{aligned}$$

All the cross terms vanish because

$$P^a{}_c \widehat{\sigma}^c{}_a = 0, \quad P^a{}_c \widehat{\omega}^c{}_a = 0, \quad \widehat{\sigma}^c{}_a \widehat{\omega}^a{}_c = 0.$$

Since $P \circ P = P$ and $T_p\mathcal{X}_\lambda$ has dimension two, we have

$$P^c_a P^a_c = P^a_a = 2.$$

Moreover,

$$\widehat{\sigma}^c{}_a \widehat{\sigma}^a{}_c = \widehat{\sigma}^{ab}\widehat{\sigma}_{ab}, \qquad \widehat{\omega}^c{}_a \widehat{\omega}^a{}_c = -\widehat{\omega}^{ab}\widehat{\omega}_{ab}.$$

Hence,

$$\widehat{B}^c{}_a \widehat{B}^a{}_c = \frac{1}{2}\theta^2 + \widehat{\sigma}^{ab}\widehat{\sigma}_{ab} - \widehat{\omega}^{ab}\widehat{\omega}_{ab}.$$

Substituting this back, we get

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \widehat{\sigma}^{ab}\widehat{\sigma}_{ab} + \widehat{\omega}^{ab}\widehat{\omega}_{ab} - R_{ab}N^aN^b$$

□

Lemma 5. *On the screen space $T_p\mathcal{X}_\lambda$, one has $\widehat{\omega}_{bc} = 0$.*

Proof. By definition, $\widehat{B}^b{}_c = B^b{}_d P^d{}_c$. Using $P^d{}_c = \delta^d{}_c + N^d W_c + W^d N_c$ and $B^b{}_d N^d = N^d \nabla_d N^b = 0$, we obtain

$$\widehat{B}^b{}_c = B^b{}_c + N_c B^b{}_d W^d.$$

Hence antisymmetrization gives $N_{[a}\widehat{\omega}_{bc]} = N_{[a}B_{bc]}$.

Write $N^\flat = g(N, \cdot)$. In the Gaussian null coordinates, $\mathcal{X} = \{r = 0\}$ and both $N^\flat$ and dr annihilate $T_p\mathcal{X}$. Thus, along $\mathcal{X}$, one has $N^\flat = \varphi\, dr$ for a nonvanishing function φ. Consequently

$$N^\flat \wedge dN^\flat = \varphi\, dr \wedge (d\varphi \wedge dr) = 0.$$

Since $B_{bc} = \nabla_c N_b$, this implies $N_{[a}\widehat{\omega}_{bc]} = 0$. Expanding and contracting with W^a, while using $g(N, W) = -1$ and the fact that $\widehat{\omega}$ is a screen tensor, yields

$$0 = -\frac{1}{3}\widehat{\omega}_{bc}.$$

Therefore $\widehat{\omega}_{bc} = 0$. □

Lemma 6.

$$\frac{d\theta}{d\lambda} \leq -\frac{1}{2}\theta^2.$$

Proof. From the previous lemma, $\widehat{\omega}_{ab} = 0$. Since h_{ij} is a Riemannian metric, choose an orthonormal basis $\{E_1, E_2\}$ of $T_p\mathcal{X}_\lambda$. Since $\widehat{\sigma}_{ab}$ is symmetric and traceless, its matrix in this basis has the form

$$(\widehat{\sigma}_{AB}) = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}.$$

Hence

$$\widehat{\sigma}^{ab}\widehat{\sigma}_{ab} = \widehat{\sigma}^{AB}\widehat{\sigma}_{AB} = 2a^2 + 2b^2 \geq 0.$$

The null energy condition gives $R_{ab}N^a N^b \geq 0$. Therefore, Raychaudhuri's equation gives

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \widehat{\sigma}^{ab}\widehat{\sigma}_{ab} + \widehat{\omega}^{ab}\widehat{\omega}_{ab} - R_{ab}N^a N^b \leq -\frac{1}{2}\theta^2.$$

□

Theorem 2. *$\theta \to -\infty$ before λ reaches $\frac{2}{|\theta_0|}$*

Proof. Along either family, the initial value satisfies $\theta(0) \leq \theta_0 < 0$. Integrating the differential inequality while θ remains finite gives $\theta(\lambda)^{-1} - \theta(0)^{-1} \geq \lambda/2$, and hence

$$\theta \leq \frac{1}{1/\theta_0 + \lambda/2}.$$

The right-hand side tends to $-\infty$ as $\lambda \uparrow 2/|\theta_0|$. □

3.2 First and Second Variation

Consider a smooth one-parameter family of curves $H : [0, L] \times (-\varepsilon, \varepsilon) \to \mathcal{M}$, and write $\gamma_s(t) = H(t, s)$. Along the variation let $N = H_*(\partial_t)$ and $J = H_*(\partial_s)$. Since ∂_t and ∂_s commute and the Levi-Civita connection is torsion free, we have $[N, J] = 0$ and hence $\nabla_J N = \nabla_N J$. We use the normalized energy $E(s) = \frac{1}{2}\int_0^L g(N, N)\, dt$.

Lemma 7. *The first variation of the energy is given by*

$$E'(0) = g(J, N)|_{t=0}^{t=L} - \int_0^L g(J, \nabla_N N)\ dt.$$

Proof. Differentiating under the integral sign and using metric compatibility gives

$$E'(s) = \int_0^L g(\nabla_J N, N)\, dt = \int_0^L g(\nabla_N J, N)\, dt.$$

The identity $\frac{d}{dt} g(J, N) = g(\nabla_N J, N) + g(J, \nabla_N N)$ and integration by parts now give the stated formula. □

If the endpoints are fixed, then $J(0) = J(L) = 0$. Thus $E'(0) = 0$ for every such variation whenever γ_0 is an affinely parametrized geodesic, because then $\nabla_N N = 0$. Conversely, the first-variation formula and variations supported in the interior show that a critical point of the energy is a geodesic.

For the boundary term relevant to a trapped surface, suppose more generally that $\gamma_s(0)$ lies on a spacelike two-surface S, that $N(0, s)$ is normal to S, and that the final endpoint $\gamma_s(L)$ is fixed. Then $J(0) \in T_{\gamma_0(0)}S$, $J(L) = 0$, and $g(J(0), N(0)) = 0$.

Lemma 8. *Assume that γ_0 is an affinely parametrized geodesic and that the variation satisfies the boundary conditions just described. The second variation of the energy is*

$$E''(0) = \int_0^L \left[g(\nabla_N J, \nabla_N J) - g(R(J, N)N, J)\right] dt + g\big(\pi_S\big(\nabla_{J(0)} N\big)\, , J(0)\big)\, .$$

where $\pi_S : T_{\gamma_0(0)}\mathcal{M} \to T_{\gamma_0(0)}S$ is the orthogonal projection.

Proof. Our curvature convention gives the commutation identity

$$\nabla_J \nabla_N N - \nabla_N \nabla_J N = R(J, N)N.$$

Using $\nabla_J N = \nabla_N J$, we obtain along γ_0 $\nabla_J \nabla_N N = \nabla_N \nabla_N J + R(J, N)N$. Differentiate the first-variation formula at $s = 0$. Since $\nabla_N N = 0$ there, and after integrating the $\nabla_N \nabla_N J$ term by parts, the terms $[g(J, \nabla_N J)]_0^L$ cancel and leave

$$E''(0) = [g(\nabla_J J, N)]_0^L + \int_0^L \big(g(\nabla_N J, \nabla_N J) - g(R(J, N)N, J)\big)\, dt.$$

At $t = L$ the first term vanishes because the endpoint is fixed. At $t = 0$, differentiate the orthogonality relation $g(J, N) = 0$ along the initial curve in S. This gives $g(\nabla_J J, N) + g(J, \nabla_J N) = 0$. Since $\nabla_J N = \nabla_N J$ at the initial point, the remaining boundary contribution is $g(\pi_S(\nabla_{J(0)} N), J(0))$, as claimed. □

If every longitudinal curve γ_s is a geodesic, differentiating $\nabla_N N = 0$ in the s direction gives the *Jacobi equation*

$$\nabla_N \nabla_N J + R(J, N)N = 0.$$

Thus the variation field of a geodesic variation is a Jacobi field. Conversely, a Jacobi field is determined by $J(0)$ and $\nabla_N J(0)$ and locally arises from a variation through geodesics.

Definition 6. *For the outward family, let $F : T^2 \times [0, L] \to \mathcal{M}$ be the normal exponential map from the Gaussian null-coordinate section. A point $z = F(p, \lambda_0)$ is a focal point of T^2 along the generator if $dF_{(p,\lambda_0)}$ is singular. Equivalently, there is a nonzero Jacobi field X with $X(0) \in T_p T^2$, $\nabla_N X(0) = \nabla_{X(0)} N$, and $X(\lambda_0) = 0$. The inward family is defined identically using G and W in place of F and N.*

Lemma 9. *Suppose that $z = \gamma_0(\lambda_0)$ is focal to T^2 and lies in the interior of a null generator segment $\gamma_0 : [0, L] \to \mathcal{M}$. Then there is a future-directed timelike curve from T^2 to $\gamma_0(L)$.*

Proof. Here a vector field is *admissible* if it is continuous and piecewise smooth along γ_0, with initial value tangent to T^2 and final value zero. For admissible vector fields V_1, V_2 along γ_0, with $V_1(0), V_2(0) \in T_{\gamma_0(0)} T^2$ and $V_1(L) = V_2(L) = 0$, define

$$I(V_1, V_2) = \int_0^L \big[g(\nabla_N V_1, \nabla_N V_2) - g(R(V_1, N)N, V_2) \big] dt + g\big(\pi_{T^2}(\nabla_{V_1(0)} N), V_2(0)\big).$$

The second-variation formula says that the second variation in direction V_1 is $I(V_1, V_1)$. Let X be the focal Jacobi field and extend it by zero from z to $\gamma_0(L)$. The Jacobi equation and integration by parts give $I(X, X) = 0$ and

$$I(X, Y) = g(\nabla_N X(\lambda_0), Y(\lambda_0))$$

for every admissible Y. Choose a smooth field Y_0 near z such that

$$g(\nabla_N X(\lambda_0), Y_0(\lambda_0)) < 0.$$

Let χ be a bump function supported near λ_0 and satisfying $\chi(\lambda_0) = 1$. For sufficiently small $\varepsilon > 0$, set $Y = \varepsilon \chi Y_0$. Then $Y(0) = Y(L) = 0$ and

$$I(X + Y, X + Y) = 2I(X, Y) + I(Y, Y) < 0.$$

Realizing $X + Y$ as a variation field pushes the null segment to the timelike side near the focal point. After smoothing, the push-up lemma gives a future-directed timelike curve from T^2 to $\gamma_0(L)$. □

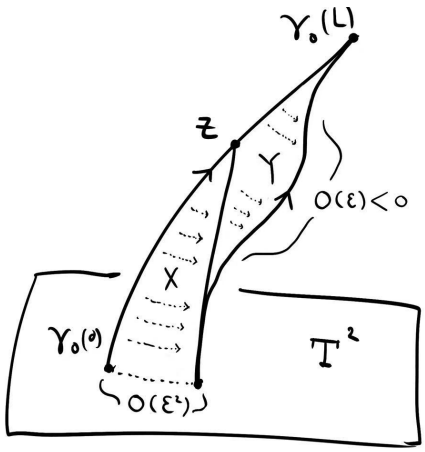


Figure 3: Intuitive idea of lemma 9

4 Topological Aspects

4.1 Causal Structure

Definition 7. *For points $p, q \in \mathcal{M}$, write $p \ll q$, if there is a future-directed timelike curve from p to q, and write $p \leq q$, if either $p = q$ or there is a future-directed causal curve from p to q. Finally, write $p < q$ when $p \leq q$ and $p \neq q$.*

Definition 8. *For a spacetime $\mathcal{M}$, a set $A \subset \mathcal{M}$ is achronal if there does not exist a timelike curve $\gamma : [a, b] \to \mathcal{M}$ with $\gamma(a) \in A,\ \gamma(b) \in A$.*

Definition 9. *For a subset $A \subset \mathcal{M}$, define*

$$I^+(A) := \{q \in \mathcal{M} : p \ll q \text{ for some } p \in A\},$$
$$J^+(A) := \{q \in \mathcal{M} : p \leq q \text{ for some } p \in A\},$$
$$E^+(A) := J^+(A) \setminus I^+(A).$$

The past sets $I^-(A)$, $J^-(A)$, and $E^-(A)$ are defined analogously. The set $E^+(A)$ is called the future horismos *of A.*

Definition 10. *If $A \subset \mathcal{M}$ is achronal, the future Cauchy development of A is the set $D^+(A)$ of all points $p \in \mathcal{M}$ such that every past-inextendible causal curve through p meets A. In particular, $A \subset D^+(A)$.. The set $D^-(A)$ is defined analogously*

Definition 11. *The Cauchy development of A is the set $D(A) = D^+(A) \cup D^-(A)$.*

4.2 Push-up lemma

Definition 12. *An open set $\mathcal{U} \subset \mathcal{M}$ is a* convex normal neighbourhood *if every pair of points in $\mathcal{U}$ is joined by a unique geodesic segment contained in $\mathcal{U}$. Equivalently, for every $x \in \mathcal{U}$, the restriction of exp_x to a star-shaped open subset of $T_x\mathcal{M}$ is a diffeomorphism onto $\mathcal{U}$.*

Definition 13. *For a future-directed causal absolutely continuous curve $\sigma : [a, b] \to \mathcal{M}$, its* Lorentzian length *is $L(\sigma) = \int_a^b \sqrt{-g(\sigma', \sigma')}\, dt$.*

Lemma 10. *Let $\mathcal{M}$ be a spacetime. Let $\mathcal{U}$ be a convex normal neighbourhood of $p \in \mathcal{M}$, and let $\sigma : [0, 1] \longrightarrow \mathcal{U}$ be any future-directed causal absolutely continuous curve starting from p.*

Then the Lorentzian length of σ is less than or equal to that of the unique future-directed causal geodesic connecting its endpoints. Equality holds precisely when the two curves differ by a monotone reparametrization.

Proof. book □

Theorem 3 (Push-up lemma)**.** *For $p, q, r \in \mathcal{M}$,*

$$\begin{aligned} p \le q,\ q \ll r \quad &\Longrightarrow \quad p \ll r, \\ p \ll q,\ q \le r \quad &\Longrightarrow \quad p \ll r. \end{aligned}$$

Proof. We prove the first implication. The second follows by reversing the time orientation. Choose a future-directed causal curve from p to q and a future-directed timelike curve from q to r, and concatenate them to a piecewise smooth causal curve $\eta : [0, 1] \to \mathcal{M}$. Its image is compact, so it is covered by finitely many convex normal neighbourhoods. After applying the Lebesgue number lemma to the inverse images of this cover, choose a partition $0 = t_0 < t_1 < \cdots < t_k = 1$ such that every segment $\eta([t_{i-1}, t_i])$ lies in one convex normal neighbourhood U_i. Refine the partition so that the last segment is timelike.

Set $x_i = \eta(t_i)$. In U_k, replace the last segment by the unique radial geodesic from x_{k-1} to $x_k = r$. The local lemma shows that this geodesic is timelike, because the segment it replaces has positive Lorentzian length. Thus $x_{k-1} \ll r$.

Suppose inductively that a timelike curve from x_i to r has already been constructed. Choose a point y on this curve sufficiently close to x_i that $y \in U_i$ and the short timelike segment from x_i to y lies in U_i. The concatenation of $\eta|_{[t_{i-1}, t_i]}$ with this short segment is a causal curve in U_i with positive Lorentzian length. Applying the local lemma inside U_i replaces it by a timelike geodesic from x_{i-1} to y. Concatenating this with the remaining timelike curve from y to r gives $x_{i-1} \ll r$.

There are only finitely many members of the partition, so the induction ends with a piecewise smooth timelike curve from $x_0 = p$ to r. Smoothing its finitely many corners preserves timelikeness. Hence $p \ll r$. □

For every subset $A \subset \mathcal{M}$, the push-up lemma gives

$$I^+\big(J^+(A)\big) = I^+(A) = J^+\big(I^+(A)\big),$$

and, with the time orientation reversed,

$$I^-\big(J^-(A)\big) = I^-(A) = J^-\big(I^-(A)\big).$$

4.3 Cauchy hypersurfaces

We first record the quasi-limit construction used below. It replaces an ordinary limit curve, which need not be piecewise smooth, by a causal broken geodesic. Fix a convex covering $\mathcal{U}$ of $\mathcal{M}$ and a sequence $\{\alpha_n\}$ of future-directed causal curves. A *limit sequence* relative to $\mathcal{U}$ is a finite or infinite sequence $p_0 < p_1 < \cdots$ with the following properties.

(L1) For every i there are a subsequence $\{\alpha_m\}$ and parameters $s_{m0} < \cdots < s_{mi}$ such that $\alpha_m(s_{mj}) \to p_j$ for $0 \le j \le i$. Moreover, for each $j < i$, the points p_j, p_{j+1} and all the curve segments $\alpha_m|_{[s_{mj}, s_{m,j+1}]}$ lie in one member of $\mathcal{U}$.

(L2) An infinite limit sequence does not converge. A finite one has at least two points and is maximal among sequences satisfying (L1).

Joining each p_i to p_{i+1} by the unique causal geodesic in the corresponding convex set gives a causal broken geodesic, called a *quasi-limit* of $\{\alpha_n\}$.

Proposition 1 (Quasi-limit construction)**.** *Suppose that $\alpha_n(0) \to p$ and that no neighbourhood of p contains the images of all but finitely many of the curves α_n. Then, relative to every convex covering, a subsequence has a limit sequence beginning at p. If all the α_n are future-inextendible, the resulting quasi-limit is future-inextendible.*

Proof. Choose a locally finite cover $\mathcal{V}$ by precompact open sets such that the closure of every member of $\mathcal{V}$ is contained in a member of $\mathcal{U}$. Select $V_0 \in \mathcal{V}$ containing $p_0 = p$ for which infinitely many α_n start in V_0 and leave it. Pass to those curves, take their first exit points from V_0, and then pass to a subsequence on which the exit points converge to some $p_1 \in \partial V_0$.

Now choose $V_1 \in \mathcal{V}$ containing p_1 and repeat, retaining curves that leave V_1, take their first exit points after entering V_1, and extract a subsequence converging to $p_2 \in \partial V_1$. Continue for as long as possible. Whenever several V_i are available, choose one that has been used the fewest times. The successive segments and their endpoint limits lie in a common member of $\mathcal{U}$, so (L1) holds. The fairness rule and local finiteness of $\mathcal{V}$ prevent an infinite sequence $\{p_i\}$ from converging, because otherwise a neighbourhood of its limit would meet only finitely many members of $\mathcal{V}$, forcing one eligible member to have been passed over infinitely often. Thus (L2) holds in the infinite case.

If the procedure stops at p_k, the remaining curves stay in the compact closure of V_k. After restricting to closed parameter intervals and taking a subsequence, their final points converge to some $q \in \overline{V_k}$. Append q when $q \neq p_k$. When $q = p_k$, the sequence is already maximal. This gives (L2) in the finite case. Finally connect successive vertices by the causal geodesics in the chosen convex sets. When the original curves are future-inextendible the construction cannot

terminate, so the vertices form a nonconvergent infinite sequence and the quasi-limit is future-inextendible. □

Definition 14. *The strong causality condition holds at $p \in \mathcal{M}$ provided that, given any neighbourhood U of p, there is a neighbourhood $V \subset U$ of p such that every causal curve segment with endpoints in V lies entirely in U. The spacetime is* strongly causal *if this condition holds at every point.*

Definition 15. *For $p, q \in \mathcal{M}$, the* causal diamond *from p to q is $J(p,q) := J^+(p) \cap J^-(q)$. A spacetime $\mathcal{M}$ is* globally hyperbolic *if it is strongly causal and $J(p,q)$ is compact for every $p \leq q$.*

Lemma 11 (Avoidance construction)**.** *Let C be closed and let $\alpha : [0,\infty) \to \mathcal{M} \setminus C$ be a past-inextendible causal curve starting at p. If p_0 can be reached from p by a future-directed timelike curve in $\mathcal{M} \setminus C$, then there is a past-inextendible timelike curve in $\mathcal{M} \setminus C$ starting at p_0.*

Proof. Work in the open spacetime $\mathcal{M} \setminus C$ and choose any metric d inducing its topology. After reparametrising α, arrange that $\{\alpha(n)\}$ does not converge. Inductively choose p_n so that $\alpha(n) \ll p_n \ll p_{n-1}$ and $d(p_n, \alpha(n)) < 1/n$. This is possible because the chronological relation is open and a point can be moved a sufficiently small amount to its timelike future. Join p_{n-1} to p_n by past-directed timelike segments. Their concatenation is a past-directed timelike curve avoiding C. The sequence $\{p_n\}$ does not converge, so the concatenated curve has no past endpoint and is therefore past-inextendible. □

Lemma 12. *Every inextendible causal curve meets a Cauchy hypersurface $\Sigma \subset \mathcal{M}$.*

Proof. The definition first implies that Σ is achronal. Moreover, $\mathcal{M} \setminus \Sigma = I^-(\Sigma) \sqcup I^+(\Sigma)$, since an inextendible timelike curve through any point meets Σ exactly once, and the point lies before or after that intersection. Hence Σ is closed.

Suppose that an inextendible causal curve α avoids Σ. Reverse time if necessary and choose a past-inextendible half of α beginning at a point $p \in I^+(\Sigma)$. Choose p_0 in the timelike future of p within $\mathcal{M} \setminus \Sigma$. The avoidance construction gives a past-inextendible timelike curve from p_0 that does not meet Σ. Extend it to the future by a future-inextendible timelike curve. Since $p_0 \in I^+(\Sigma)$, this future extension remains in $I^+(\Sigma)$. We have therefore produced an inextendible timelike curve missing Σ, contrary to the definition of a Cauchy hypersurface. □

Lemma 13. *Let $A \subset \mathcal{M}$ be an achronal set, then* $\mathrm{int}(D(A))$ *is a globally hyperbolic set*

Proof. We indicate where the quasi-limit construction enters. First, there are no closed causal curves in $D(A)$, because repeated traversal of such a curve would give an inextendible causal curve meeting the achronal set A more than once.

Suppose strong causality fails at $p \in \operatorname{int} D(A)$. Then there are causal segments α_n whose two endpoints converge to p but which all leave one fixed neighbourhood of p. The quasi-limit construction produces a limit sequence from p. A finite limit sequence would return to p and hence give $p < p$, which has just been excluded. Thus it produces a future-inextendible quasi-limit from p. Applying the construction to the same segments with the time direction reversed produces a past-inextendible quasi-limit. Every inextendible causal curve through an interior point of $D(A)$ must cross from $I^-(A)$ to $I^+(A)$. This follows directly from the definition of $D(A)$, with the avoidance construction used if one of its halves only approaches A. The two quasi-limits therefore force, for all sufficiently large n, a single causal segment α_n to meet both $I^+(A)$ and $I^-(A)$ in the wrong order. The push-up lemma then gives two timelike-related points of A, contradicting achronality. Hence strong causality holds.

It remains to prove compactness of the causal diamonds. Let $p < q$ lie in $\operatorname{int} D(A)$ and take a sequence $x_n \in J^+(p) \cap J^-(q)$. Choose a causal curve α_n from p through x_n to q. A finite limit sequence from p ends at q. By (L1), infinitely many x_n lie in one of the finitely many convex pieces between two successive vertices, and the compact closure of that piece supplies a convergent subsequence, and closedness of the local causal relation places its limit in $J^+(p) \cap J^-(q)$. If every limit sequence from p were infinite, its future-inextendible quasi-limit and the time-reversed construction from q would again cross $I^+(A)$ and $I^-(A)$ and contradict achronality. Thus the finite alternative must occur, proving sequential compactness and hence compactness. The same crossing argument shows that the diamond remains in $\operatorname{int} D(A)$. Therefore $\operatorname{int} D(A)$ is globally hyperbolic. □

Theorem 4. *A spacetime $\mathcal{M}$ with a Cauchy hypersurface $\Sigma \subset \mathcal{M}$ is globally hyperbolic.*

Proof. The hypersurface Σ is achronal by definition. By the preceding lemma every inextendible causal curve meets Σ, so $D(\Sigma) = \mathcal{M}$. Since $\operatorname{int}(\mathcal{M}) = \mathcal{M}$, the preceding result shows that $\mathcal{M}$ is globally hyperbolic. □

Proposition 2. *Fix a smooth future-directed timelike vector field U on $\mathcal{M}$. Projection along its flow defines a continuous map $\rho : \mathcal{M} \to \Sigma$ fixing Σ.*

Proof. Each maximal integral curve of U meets Σ once since Σ is Cauchy. The flow is transverse to Σ, so the implicit function theorem gives smooth dependence of the intersection on the initial point. The projection fixes Σ. □

4.4 Topology of $E^+(T^2)$

Lemma 14. *$E^+(T^2)$ is achronal.*

Proof. If not say $q, p \in E^+(T^2)$ with $q \ll p$, then by the push-up lemma, there exists $x \in T^2$ such that $x \ll p$, contradiction. □

Lemma 15. *$E^+(T^2)$ is a closed subset of $\mathcal{M}$*

Proof. Let $q_n \in E^+(T^2)$ with $q_n \to q$. For each n, choose $p_n \in T^2$ and a future-directed causal curve α_n from p_n to q_n. Since T^2 is compact, after passing to a subsequence we may assume $p_n \to p \in T^2$. If $p = q$, then $q \in T^2 \subset J^+(T^2)$, so suppose $p \neq q$.

Choose $p_- \ll p$ and $q \ll q_+$. For all sufficiently large n, the push-up lemma gives $p_- \ll p_n$ and $q_n \ll q_+$. Hence every α_n is contained in the causal diamond $J^+(p_-) \cap J^-(q_+)$. This diamond is compact because $\mathcal{M}$ is globally hyperbolic. Apply the quasi-limit construction from the Cauchy-hypersurface section to $\{\alpha_n\}$, using a locally finite convex cover. Its limit sequence cannot be infinite, since the corresponding inextendible quasi-limit would be imprisoned in the compact diamond, whereas strong causality forbids an inextendible causal curve from being totally imprisoned, meaning that its image is contained in one compact set. Thus the limit sequence is finite. Its first vertex is p, and maximality together with $q_n \to q$ forces its last vertex to be q. Joining successive vertices gives a causal broken geodesic from p to q, so $q \in J^+(T^2)$.

Finally, $q \notin I^+(T^2)$. Otherwise the openness of $I^+(T^2)$ would imply $q_n \in I^+(T^2)$ for all sufficiently large n, contrary to $q_n \in E^+(T^2)$. Therefore $q \in E^+(T^2)$, proving closedness. □

Let $\mathcal{X}, \mathcal{Y}$ be the two null hypersurfaces from the Gaussian null-coordinate section. Using the same normal exponential-map notation, let their maximal future null generators be

$$\gamma_p(\lambda) = F(p, \lambda) = \exp_p(\lambda N_p), \qquad \beta_p(\lambda) = G(p, \lambda) = \exp_p(\lambda W_p),$$

and define

$$\overline{\mathcal{X}} = \bigcup_{p \in T^2} \mathrm{Im}(\gamma_p), \qquad \overline{\mathcal{Y}} = \bigcup_{p \in T^2} \mathrm{Im}(\beta_p).$$

Lemma 16.

$$E^+(T^2) \subset \overline{\mathcal{X}} \cup \overline{\mathcal{Y}}.$$

Proof. Let $q \in E^+(T^2)$. Since $q \in J^+(T^2)$, there is a future-directed causal curve α from some $p \in T^2$ to q. Cover the compact image of α by finitely many convex normal neighbourhoods and straighten its successive pieces as in the proof of the push-up lemma. If a piece were timelike, or if two successive null geodesic pieces formed a genuine corner, the local comparison result followed by push-up would give $p \ll q$, contrary to $q \notin I^+(T^2)$. Hence α is, up to reparametrization, a single future-directed null geodesic γ from p to q.

It remains to show that $\gamma'(0)$ is normal to T^2. Let $V \in T_pT^2$, choose a curve $p(s)$ in T^2 with $p'(0) = V$, and fix a small $\delta > 0$ such that p and $\gamma(\delta)$ lie in one convex normal neighbourhood. Vary the unique local geodesic from $p(s)$ to $\gamma(\delta)$ while keeping its final endpoint fixed. The first-variation formula gives

$$E'(0) = -g(V, \gamma'(0)).$$

If this number were nonzero, choosing the sign of s would make the nearby geodesic timelike. It could then be concatenated with the remaining null segment from $\gamma(\delta)$ to q, and the push-up lemma would again give a timelike curve

from T^2 to q. Therefore $g(V, \gamma'(0)) = 0$ for every $V \in T_pT^2$. The two future null directions normal to T^2 are N_p and W_p, so γ is one of the generators γ_p or β_p. Thus $q \in \overline{\mathcal{X}} \cup \overline{\mathcal{Y}}$. □

Lemma 17. *No generator segment from T^2 to a point of $E^+(T^2)$ contains a focal point in its interior.*

Proof. Otherwise the focal-point variation lemma from the preceding section would give a future-directed timelike curve from T^2 to that endpoint, placing it in $I^+(T^2)$ rather than $E^+(T^2)$. □

Lemma 18. *Assume that every future-directed normal null geodesic starting from T^2 extends beyond affine parameter $2/|\theta_0|$. Then $E^+(T^2)$ is compact.*

Proof. Set $L_0 = 2/|\theta_0|$. By the assumed extendibility, the two normal exponential maps

$$F(p, \lambda) = \exp_p(\lambda N_p), \qquad G(p, \lambda) = \exp_p(\lambda W_p)$$

are defined on $T^2 \times [0, L_0]$. Since T^2 is compact, the set

$$K = F\big(T^2 \times [0, L_0]\big) \cup G\big(T^2 \times [0, L_0]\big)$$

is compact.

We claim that $E^+(T^2) \subset K$. Indeed, the generator result above shows that every $q \in E^+(T^2)$ has the form $F(p, \lambda)$ or $G(p, \lambda)$ for some $p \in T^2$ and $\lambda \geq 0$. Along either generator the Raychaudhuri estimate gives $\theta \to -\infty$ at or before affine parameter L_0. Since $\frac{d}{d\lambda} \log \sqrt{h} = \theta$, the transverse area density vanishes there. Equivalently, the corresponding normal exponential map loses rank and a focal point occurs at some $\lambda_f \leq L_0$.

If the parameter of q satisfied $\lambda > L_0$, then this focal point would lie in the interior of the generator segment from T^2 to q. This is impossible by the preceding no-focal-point lemma. Therefore $\lambda \leq L_0$, proving $q \in K$.

Finally, $E^+(T^2)$ is closed in $\mathcal{M}$ by Lemma 15. Hence it is a closed subset of the compact set K, and is therefore compact. □

Lemma 19. *The set $E^+(T^2)$ is an embedded three dimensional topological manifold.*

Proof. □

4.5 Invariance of Domain

Theorem 5 (Invariance of domain)**.** *Let X, Y be topological manifolds of the same dimension. If $f : X \to Y$ is continuous and injective, then $f : X \to f(X)$ is a homeomorphism.*

By the definition of a topological manifold, it suffices to show the following special case.

Theorem 6. *Let $U \subset \mathbb{R}^n$ be open. If $f : U \to \mathbb{R}^n$ is continuous and injective, then $f(U)$ is open and $f : U \to f(U)$ is a homeomorphism.*

Proof. The standard proof uses Alexander duality. Write $\widetilde{H}_q$ and $\widetilde{H}^q$ for reduced homology and cohomology with coefficients in a fixed ring R. If $A \subset S^n$ is nonempty, compact, and locally contractible, Alexander duality gives

$$\widetilde{H}_q(S^n \setminus A, R) \cong \widetilde{H}^{n-q-1}(A, R), \qquad q \geq 0.$$

which I have no idea how to prove. □

5 Punch line

The restriction $\rho|_{E^+(T^2)}$ is continuous. It is also injective because two distinct points on the same integral curve of U are timelike related, which is impossible because $E^+(T^2)$ is achronal. Both $E^+(T^2)$ and Σ are three-dimensional topological manifolds, so invariance of domain shows that $\rho(E^+(T^2))$ is open in Σ and that $E^+(T^2) \cong \rho(E^+(T^2))$.

The set $E^+(T^2)$ is compact, hence its image is closed in Σ. Since ρ fixes Σ, it maps the connected spacetime $\mathcal{M}$ onto Σ. Thus $\rho(E^+(T^2)) = \Sigma$, so $\mathrm{E}^+(T^2) \cong \Sigma$. This is impossible because $E^+(T^2)$ is compact, whereas Σ is non-compact. The assumption of future null geodesic completeness must therefore fail.